

Assistant	CookSim				VHSim				ScreenSim			
	In-time success	Action efficiency	Conversation efficiency	Detection F1	In-time success	Action efficiency	Conversation efficiency	Detection F1	In-time success	Action efficiency	Conversation efficiency	Detection F1
Frontier API models												
Gemini 3.8 Flash	31 ± 2	31	75	<u>30</u>	21 ± 6	15	80	<u>68</u>	79 ± 2	78	60	<u>96</u>
GPT-6 Astra	26 ± 1	34	83	34	16 ± 4	14	90	58	90 ± 2	77	<u>66</u>	98
Gemini 3.1 Pro	21 ± 2	36	72	25	18 ± 3	13	<u>86</u>	75	<u>86</u> ± 2	80	29	98
Real-time streaming API models												
GPT-Realtime 2.1 (polled)	<u>27</u> ± 2	40	1	22	19 ± 5	16	81	56	67 ± 2	78	5	98
Gemini 3.8 Live (polled)	22 ± 2	30	<u>85</u>	22	18 ± 3	25	80	22	72 ± 3	78	24	94
GPT-Realtime 2.1 (native)	21 ± 8	43	45	7	—	—	—	—	—	—	—	—
Gemini 3.8 Live (native)	12 ± 5	<u>48</u>	90	1	—	—	—	—	—	—	—	—
Open-weight non-streaming VLMs												
Muse Glimmer	24 ± 1	33	2	6	16 ± 2	17	73	20	44 ± 2	65	3	89
Qwen3.8-27B	18 ± 1	37	35	17	25 ± 8	22	85	32	68 ± 2	78	13	<u>96</u>
Kimi-VL-A3B-Thinking	15 ± 1	35	31	17	<u>24</u> ± 4	25	82	65	10 ± 1	72	19	<u>56</u>
InternVideo3-8B	14 ± 2	38	64	6	18 ± 4	34	81	0	0 ± 1	84	10	32
Molmo2-8B	13 ± 1	39	2	2	3 ± 2	87	70	1	1 ± 1	87	0	16
GLM-4.6V-Flash	12 ± 1	39	75	4	20 ± 2	26	82	21	0 ± 0	81	3	28
Open-weight streaming VLMs												
Qwen3-Omni-30B	26 ± 2	43	0	11	13 ± 2	<u>46</u>	65	3	15 ± 2	74	1	61
MiniCPM-o 4.5	14 ± 1	65	0	20	15 ± 2	26	59	14	3 ± 1	76	2	45
Proact-VL	11 ± 1	<u>48</u>	71	0	20 ± 3	35	63	0	0 ± 0	92	0	1
StreamingVLM	11 ± 2	45	78	0	19 ± 1	14	80	0	0 ± 0	<u>91</u>	88	1
Reference conditions												
No assistance	—	—	—	—	15 ± 5	11	100	0	0 ± 0	84	100	0
Scripted reference	—	—	—	—	—	—	—	—	90 ± 1	78	74	95

Table 1 Task performance. Task success, execution efficiency, and error detection across three environments. Results summarize three independent runs. Scores are on a 0–100 scale; success includes run SD. Within each family, models are ordered by CookSim success. Bold and underline mark the highest and second-highest assistant values in each metric column, with ties at displayed precision. Reference conditions are not ranked; dashes indicate unavailable results.